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PROBLEMS SUBMIT CODE MY SUBMISSIONS STATUS HACKS ROOM STANDINGS CUSTOM INVOCATION

F. Machine Learning

time limit per test: 4 seconds
memory limit per test: 512 megabytes

You come home and fell some unpleasant smell. Where is it coming from?

You are given an array a . You have to answer the following queries:

- You are given two integers l and r . Let c_i be the number of occurrences of i in $a_{l:r}$, where $a_{l:r}$ is the subarray of a from l -th element to r -th inclusive. Find the **Mex** of $\{c_0, c_1, ..., c_{10^9}\}$
- You are given two integers p to x . Change a_p to x .

The **Mex** of a multiset of numbers is the smallest non-negative integer **not in** the set.

Note that in this problem all elements of a are positive, which means that $c_0 = 0$ and 0 is never the answer for the query of the second type.

Input

The first line of input contains two integers n and q ($1 \leq n, q \leq 100\,000$) — the length of the array and the number of queries respectively.

The second line of input contains n integers — $a_1, a_2, ..., a_n$ ($1 \leq a_i \leq 10^9$).

Each of the next q lines describes a single query.

The first type of query is described by three integers $t_i = 1, l_i, r_i$, where $1 \leq l_i \leq r_i \leq n$ — the bounds of the subarray.

The second type of query is described by three integers $t_i = 2, p_i, x_i$, where $1 \leq p_i \leq n$ is the index of the element, which must be changed and $1 \leq x_i \leq 10^9$ is the new value.

Output

For each query of the first type output a single integer — the **Mex** of $\{c_0, c_1, ..., c_{10^9}\}$.

Example

input	Copy
10 4 1 2 3 1 1 2 2 2 9 9 1 1 1 1 2 8 2 7 1 1 2 8	
output	Copy
2 3 2	

Note

The subarray of the first query consists of the single element — 1.

The subarray of the second query consists of four 2s, one 3 and two 1s.

The subarray of the fourth query consists of three 1s, three 2s and one 3.

Codeforces Round 466 (Div. 2)

Finished

Practice

→ Virtual participation

Virtual contest is a way to take part in past contest, as close as possible to participation on time. It is supported only ICPC mode for virtual contests. If you've seen these problems, a virtual contest is not for you - solve these problems in the archive. If you just want to solve some problem from a contest, a virtual contest is not for you - solve this problem in the archive. Never use someone else's code, read the tutorials or communicate with other person during a virtual contest.

Start virtual contest

→ Clone Contest to Mashup

You can clone this contest to a mashup.

Clone Contest

→ Submit?

Language: GNU G++23 14.2 (64 bit, ms)

Choose file: Choose File No file chosen

Submit

→ Last submissions

Submission	Time	Verdict
363923600	Feb/21/2026 22:40	Accepted
363922652	Feb/21/2026 22:29	Accepted
363922359	Feb/21/2026 22:26	Time limit exceeded on test 11
363921882	Feb/21/2026 22:22	Time limit exceeded on test 3
363921670	Feb/21/2026 22:20	Runtime error on test 2
363921404	Feb/21/2026 22:17	Runtime error on test 2
363921239	Feb/21/2026 22:16	Runtime error on test 2
363921159	Feb/21/2026 22:15	Runtime error on test 2
363921059	Feb/21/2026 22:14	Runtime error on test 2
363874245	Feb/21/2026 18:30	Time limit exceeded on test 4

→ Problem tags

brute force data structures *2600

No tag edit access